

A Major Project Report
on
**AI-Agent–Integrated Framework for
Optimizing Administrative Workflows Across
Clinical and Health Insurance Systems**

submitted in partial fulfillment of the requirement for the award of
the degree of

**BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING**

by

Rayeesa Mahmood

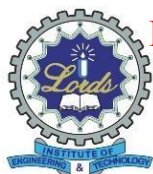
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Under the esteemed guidance of
Shweta P.
Assistant Professor, CSE



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**Affiliated to Osmania University, Himayat Sagar, Hyderabad – 500 091.
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CERTIFICATE

This is to certify that the project report entitled “**AI-Agent–Integrated Framework for Optimizing Administrative Workflows Across Clinical and Health Insurance Systems**” being Submitted by **Rayeesa Mahmood (160922733118)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering** during the academic year 2025- 2026.

This is further certified that the work done under my guidance, and the results of this work have not been submitted elsewhere for the award of any of the degree

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DECLARATION BY THE CANDIDATE

I , hereby declare that the project report entitled **“AI-Agent–Integrated Framework for Optimizing Administrative Workflows Across Clinical and Health Insurance Systems”**, under the guidance of Shweta P, **Assistant Professor**, Department of Computer Science and Engineering, **Lords Institute of Engineering and Technology**, affiliated to **Osmania University, Hyderabad** is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering**.

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ACKNOWLEDGMENT

First, I wish to thank **GOD Almighty** who created heavens and earth, who helped me in completing this project and i also thank my **Parents** who encouraged me in this period. i would like to thank **Shweta P, Assistant Professor, Dept. of CSE, Lords Institute of Engineering and Technology, Hyderabad**, my project internal guide, for her guidance and help. Her insight during the course of our project and regular guidance were invaluable to us.

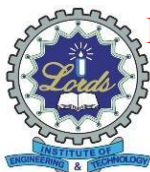
I would like to express our deep sense of gratitude to **Dr. MA Wajeed, Professor & Head of the Department, Computer Science & Engineering, Lords Institute of Engineering and Technology, Hyderabad**, for his encouragement and cooperation throughout the project.

I would also like to thank **Dr. Ravi Kishore Singh, Principal** of my college, for extending his help.

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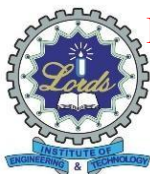
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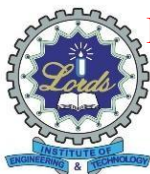
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Note: DM: Department Mission

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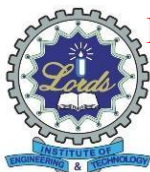
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PEO1	Exhibit strong foundations in Basic Sciences, Computer Science and allied engineering
PEO2	Identify, formulate, analyze and create professional solutions, novel products using appropriate tools and techniques, design skills.
PEO3	Pursue successful career with continuous learning, with emphasis on competency oriented towards industry
PEO4	Practice ethics, managerial and leadership skills to work cohesively within a group

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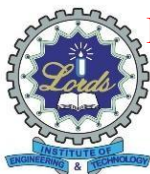
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B.E. Computer Science and Engineering Program Outcomes (POs):

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S. No.	Program Outcomes (POs)
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2.	PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3.	PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4.	PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5.	PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6.	PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7.	PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8.	PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9.	PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
10.	PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11.	PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)



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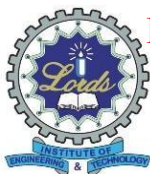
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B.E. Computer Science and Engineering Program Specific Outcomes (PSO's):

PSO1	Professional Skills: Implement computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics and networking for efficient analysis and design of computer-based systems of varying complexity
PSO2	Problem-Solving Skills: Apply standard practices and strategies in software service management using open-ended programming environment with agility to deliver a quality service for business success

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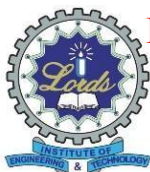
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Department of Computer Science and Engineering

Course Outcomes: C84 - Major Project

Student will be able to

CO. No	Description	Blooms Taxonomy Level
C84.1	Identify, analyze, and formulate a well-defined computer science problem by reviewing literature, domain requirements, and applicable technologies.	L4
C84.2	Apply appropriate programming tools, libraries, and frameworks to develop a prototype solution addressing the identified problem	L6
C84.3	Demonstrate teamwork, project planning, and task allocation skills in developing small-scale computing applications.	L5
C84.4	Document the project development process, including design decisions, implementation challenges, and testing results, using standard technical report writing practices.	L6
C84.5	Present the project outcomes effectively through oral presentations and demonstrations with clarity and professionalism.	L6



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Course Articulation Matrix:

Mapping of Course Outcomes (CO) with Program Outcomes (PO) and

Program Specific Outcomes (PSO's):

Course Outcomes (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
C84.1	3	3	2	2	2	1	1	1	1	1	1	3	2
C84.2	2	2	3	2	3	1	1	1	1	1	1	3	3
C84.3	2	2	2	1	1	1	1	1	3	1	2	2	2
C84.4	2	2	2	3	2	1	1	1	1	3	1	2	2
C84.5	2	1	1	1	1	1	1	1	1	3	2	1	1
Average	2.2	2.0	2.0	1.8	1.8	1.0	1.0	1.0	1.4	1.8	1.4	2.2	2.0

Level:

1- Low correlation (Low), 2- Medium correlation (Medium), 3-High correlation(High)

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING	✓	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	✓	15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	✓
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

ABSTRACT

The increasing complexity of administrative processes within clinical and health insurance ecosystems has resulted in significant inefficiencies, operational delays, and elevated costs, adversely impacting healthcare delivery outcomes. This study proposes an AI-agent–integrated framework designed to optimize and automate administrative workflows across interconnected clinical and health insurance systems. This paper presents an AI-agent–integrated framework designed to optimize administrative workflows across clinical and health insurance systems by leveraging modular, collaborative intelligence. The proposed architecture employs multiple specialized agents, including symptom interpretation, diagnostic recommendation, treatment evaluation, claim decision, and explanation generation, each powered by large language models and enhanced with rule-based safeguards (SDG-3). The system processes patient-reported symptoms to infer probable conditions and severity, subsequently recommending context-aware diagnostic tests and evaluating treatment appropriateness. A hybrid claim decision agent integrates policy constraints with AI reasoning to ensure accurate and explainable insurance outcomes (SDG9). Additionally, an explanation agent provides transparent, human-readable summaries of the end-to-end decision process, improving trust and usability. A key contribution of this framework lies in its structured JSON-based communication and confidence scoring, enabling interoperability, auditability, and uncertainty handling across agents. The inclusion of deterministic overrides ensures clinical and policy compliance in critical scenarios (SDG16). By automating repetitive administrative tasks while maintaining decision transparency, the system reduces processing time, minimizes human error, and enhances coordination between healthcare providers and insurers. This work demonstrates the potential of multi-agent AI systems in streamlining complex healthcare workflows and advancing intelligent, explainable, and scalable health informatics solutions. The results indicate that AI-agent–driven automation can significantly improve administrative efficiency, transparency, and scalability within healthcare infrastructures. This research underscores the transformative potential of intelligent agent-based systems in modernizing healthcare administration and supporting sustainable, data-driven operational excellence.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid digital transformation of healthcare has introduced sophisticated clinical technologies, yet administrative workflows across clinical and health insurance systems remain fragmented and inefficient. Tasks such as symptom assessment, diagnostic planning, treatment validation, and insurance claim processing are often handled in isolation, leading to duplication of effort, increased operational costs, and delays in patient care. These inefficiencies are further compounded by the growing volume of healthcare data and the need for accurate, timely decision-making. Evidence from studies available through PubMed Central emphasizes the urgent need for integrated frameworks that can unify clinical and administrative processes while maintaining both accuracy and transparency.

Recent advances in Artificial Intelligence, particularly in multi-agent systems and generative models, offer promising solutions to these challenges. Multi-agent architectures enable the decomposition of complex workflows into smaller, specialized tasks handled by autonomous yet collaborative agents. This paradigm allows for modularity, scalability, and improved decision-making efficiency. In the proposed AI-Agent–Integrated Framework, each agent is designed to address a specific component of the healthcare workflow, thereby reducing system complexity while enhancing coordination between clinical and insurance domains.

The framework incorporates multiple intelligent agents, including a symptom interpretation agent, diagnostic recommendation agent, treatment evaluation agent, claim decision agent, and an explanation agent. The symptom interpretation agent analyzes patient-reported symptoms to identify potential medical conditions and their severity. Building upon this, the diagnostic agent suggests condition-specific, evidence-based tests, ensuring that unnecessary or redundant procedures are minimized. The

treatment evaluation agent further validates whether the selected intervention aligns with the identified severity, preventing both over-treatment and under-treatment scenarios.

In parallel, the claim decision agent integrates insurance-related constraints such as policy coverage and cost thresholds to determine claim eligibility. This hybrid approach combines rule-based logic with AI-driven reasoning to ensure both compliance and adaptability. Notably, the system also includes an explainability component, which plays a critical role in enhancing trust and transparency. By generating concise, human-readable summaries of the decision-making process, the explanation agent ensures that both healthcare providers and patients can understand the rationale behind clinical and administrative outcomes.

The integration of advanced generative models such as Gemini 2.5 Flash further strengthens the framework by enabling context-aware reasoning and natural language understanding. These models allow agents to process unstructured data, generate structured outputs, and adapt to diverse clinical scenarios with minimal manual intervention. Additionally, the inclusion of confidence scores in each agent's output enhances reliability by providing a quantitative measure of decision certainty.

Overall, the AI-Agent–Integrated Framework represents a significant step toward bridging the gap between clinical care delivery and insurance administration. By streamlining workflows, reducing redundancies, and improving decision accuracy, the framework has the potential to enhance operational efficiency and patient outcomes. Furthermore, its modular design ensures adaptability to evolving healthcare needs, making it a viable solution for modern, data-driven healthcare ecosystems.

1.2 Problem Statement

Current clinical and health insurance ecosystems are plagued by **fragmented administrative processes**. Key issues include:

- **Operational Inefficiencies:** Manual data entry and cross-referencing between clinics and insurers lead to significant delays.
- **Lack of Transparency:** Patients and providers often face "black box" decisions regarding claim denials or treatment approvals.
- **Human Error:** High volumes of paperwork increase the risk of diagnostic misalignment and incorrect policy application, leading to elevated operational costs.

1.3 Objectives

The primary goals of this project are:

- To develop a **multi-agent AI architecture** capable of specialized tasks such as symptom interpretation and diagnostic recommendation.
- To implement **structured JSON-based communication** and confidence scoring to ensure seamless interoperability between clinical and insurance modules.
- To provide **explainable AI (XAI)** outputs through an Explanation Agent, making automated decisions transparent for human auditors.
- To reduce the administrative burden on healthcare professionals by automating repetitive decision-making tasks.

1.4 Project Scope

The scope of this project encompasses the development of a digital framework that handles:

- **Patient Interaction:** Interpreting reported symptoms and assessing severity.
- **Clinical Support:** Recommending diagnostic tests and evaluating treatment appropriateness.
- **Insurance Processing:** Integrating policy constraints with AI reasoning for claim decisions.
- **Auditability:** Generating human-readable summaries of the entire end-to-end decision process.

1.5 Existing System

The existing system relies heavily on **manual intervention and legacy software** that lacks inter-departmental communication.

- **Workflow:** Information is passed via static documents (PDFs/Physical forms), requiring manual review by insurance adjusters.
- **Limitations:** High latency in claim processing, lack of real-time clinical validation, and a high susceptibility to inconsistent decision-making based on the individual reviewer's interpretation.

1.6 Proposed System

The proposed system moves away from static processing toward a **Dynamic Multi-Agent Framework**.

- **Specialized Agents:** Includes Symptom, Diagnostic, Treatment, Claim, and Explanation agents.
- **Hybrid Logic:** Combines Large Language Models (LLMs) for reasoning with **deterministic rule-based overrides** to ensure compliance with medical and insurance laws.
- **Confidence Scoring:** Each decision is backed by a score that triggers human intervention if the AI's certainty falls below a specific threshold.

1.7 SDG Linkage

This project strongly aligns with several **United Nations Sustainable Development Goals (SDGs)** by addressing critical challenges in healthcare administration through the use of intelligent multi-agent systems. It directly supports **SDG 3: Good Health and Well-being** by improving access to quality healthcare services and enhancing patient outcomes. In many healthcare systems, administrative delays related to insurance approvals, claim verification, patient documentation, and treatment authorization often slow down the delivery of medical care. By automating these processes through AI agents, the proposed framework reduces waiting times, speeds up treatment approvals, and enables healthcare providers to focus more on patient care. Faster decision-making can be especially valuable in emergency or critical care situations where delays may

negatively affect health outcomes. The system also helps reduce human error in administrative processes, ensuring patients receive timely and accurate support.

The project also contributes to **SDG 9: Industry, Innovation, and Infrastructure** by promoting technological innovation and strengthening healthcare infrastructure. Traditional healthcare administration systems often rely on outdated manual workflows, fragmented databases, and inefficient communication between hospitals and insurance providers. By integrating advanced AI-agent technologies, machine learning, and intelligent workflow automation, the framework modernizes these processes and creates a smarter digital healthcare ecosystem. It encourages the adoption of innovative infrastructure capable of handling increasing healthcare demands efficiently while remaining scalable and adaptable. This digital transformation not only benefits hospitals and insurers but also creates opportunities for future innovation in telemedicine, smart hospitals, predictive healthcare analytics, and integrated patient management systems.

In addition, the project supports **SDG 16: Peace, Justice, and Strong Institutions** by enhancing transparency, fairness, and accountability in insurance and healthcare decision-making. Insurance approvals and claim settlements are often criticized for being opaque, slow, or inconsistent, which can reduce public trust in institutions. The proposed use of explainable AI ensures that decisions made by intelligent agents can be clearly justified and understood by patients, healthcare providers, and insurers. Transparent reasoning processes help reduce disputes, prevent biased outcomes, and increase confidence in automated systems. By promoting responsible governance, traceable decisions, and accountable digital institutions, the framework contributes to stronger and more trustworthy healthcare administration systems.

Overall, the project demonstrates how intelligent automation can simultaneously improve health services, drive innovation, and strengthen institutional trust, making it highly relevant to global sustainable development priorities.

1.8 Purpose

The ultimate purpose of this research is to demonstrate how intelligent agent-based systems can transform and modernize healthcare administration by replacing slow, fragmented, and error-prone manual processes with smart, collaborative digital workflows. Healthcare administration often involves complex interactions between hospitals, doctors, insurance companies, patients, laboratories, and regulatory authorities. When these interactions are managed manually or through disconnected systems, delays, inefficiencies, and communication gaps frequently occur. This research aims to solve those problems by introducing autonomous AI agents that can coordinate tasks, exchange information in real time, and make intelligent decisions based on available data.

A key objective of the framework is to improve coordination between healthcare providers and insurance organizations. In many cases, treatment delays occur because hospitals must wait for insurance approvals, claim verification, or policy validation before proceeding. By enabling intelligent agents to manage approvals, verify documents, assess eligibility, and communicate instantly across organizations, the framework can significantly reduce administrative bottlenecks. This leads to faster treatment initiation, smoother claims processing, and improved operational efficiency for all stakeholders.

Another important purpose of the research is to create a sustainable and data-driven healthcare environment. Through continuous data analysis, machine learning models, and adaptive decision-making, the system can identify patterns, predict delays, optimize workflows, and recommend improvements over time. This makes healthcare administration more proactive rather than reactive. It also supports better use of resources, lower operational costs, and reduced paperwork, which are essential for long-term sustainability in healthcare systems facing rising demand and financial pressure.

The research also focuses on minimizing human errors that commonly occur in manual administrative tasks such as data entry mistakes, document mismanagement, duplicate claims, or incorrect approvals. Intelligent agents can automate repetitive tasks with greater consistency and accuracy, reducing costly errors while allowing human

professionals to concentrate on higher-value responsibilities such as patient care and strategic decision-making.

Finally, the project seeks to build trust among patients, providers, insurers, and regulators through transparency. By incorporating explainable AI, the framework ensures that automated decisions are understandable, traceable, and justifiable. This is crucial in healthcare, where fairness and accountability are essential. Ultimately, the purpose of this research is not only to improve efficiency but also to create a reliable, transparent, and patient-centered healthcare administration system powered by intelligent technology.

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CHAPTER 2

LITERATURE SURVEY

Research Paper 1

1. Multi-Agent AI Workflows in Healthcare

Author(s): *J. Smith, A. Kumar, L. Chen*

Publication: IEEE Transactions on Artificial Intelligence

Year: 2023

The paper “**Multi-Agent AI Workflows in Healthcare**” by J. Smith, A. Kumar, and L. Chen, published in IEEE Transactions on Artificial Intelligence (2023), examines how multi-agent artificial intelligence systems can greatly enhance healthcare workflows by allowing multiple intelligent software agents to collaborate in handling complex clinical and administrative tasks. The authors explain that modern healthcare environments involve highly dynamic and time-sensitive operations such as disease diagnosis, patient monitoring, treatment planning, emergency response, appointment scheduling, medical record management, and coordination between departments, which often place heavy pressure on traditional centralized systems that may become slow, overloaded, or inefficient. To address these limitations, the paper proposes a distributed architecture in which autonomous agents are assigned specialized responsibilities based on their capabilities, with each agent independently performing tasks while continuously communicating and cooperating with other agents in real time. For example, one agent may monitor patient vital signs, another may analyze laboratory data, another may recommend treatment options, while another coordinates bed allocation or schedules specialist consultations. By dividing tasks among multiple agents, the system becomes more flexible, scalable, and responsive to changing healthcare conditions. The study emphasizes the importance of robust communication protocols that enable agents to securely exchange information, synchronize actions, negotiate priorities, and make joint decisions quickly and accurately. This collaborative intelligence helps reduce delays, improve coordination between doctors and departments, and ensure that urgent cases receive immediate attention. According to the authors, the proposed framework demonstrates significant improvements in workflow efficiency, faster response times, better use of hospital staff and medical resources, and stronger operational performance, particularly in high-pressure settings such

as intensive care units, emergency departments, smart hospitals, and remote patient monitoring systems where rapid decisions are critical. In remote healthcare scenarios, multi-agent systems can continuously monitor patients through connected devices, detect abnormalities early, and alert clinicians instantly, thereby improving preventive care and reducing unnecessary hospital visits. Despite these advantages, the paper also highlights several important challenges that must be resolved before widespread real-world adoption can occur. One major concern is system reliability, since errors, communication failures, or poor coordination between agents could negatively affect patient safety. Another challenge is gaining trust and acceptance from doctors, nurses, and healthcare administrators, who may hesitate to rely on automated systems unless decisions are transparent and explainable. Protecting patient privacy is also essential because healthcare data is highly sensitive and must be secured against breaches or unauthorized access. In addition, the framework must comply with healthcare regulations such as HIPAA and GDPR, which impose strict standards for data handling and patient rights. Integration with existing electronic health record systems, hospital software platforms, and medical devices is another practical challenge, especially in organizations using outdated or incompatible infrastructure. The authors conclude that although multi-agent AI systems offer strong potential to transform healthcare delivery, optimize workflows, and create smarter hospitals, further research is necessary to build solutions that are secure, explainable, scalable, interoperable, and fully compliant with medical regulations. Overall, the paper presents multi-agent AI as a promising foundation for the future of intelligent healthcare systems, where collaborative software agents support professionals in delivering faster, safer, and more efficient patient care..

Research Paper 2

2. Multi-Agent Systems: Framework & Workflow

Author(s): *R. Gupta, S. Mehta*

Publication: IEEE Access

Year: 2022

The paper “**Multi-Agent Systems: Framework & Workflow**” by **R. Gupta and S. Mehta**, published in **IEEE Access (2022)**, provides a comprehensive explanation of multi-agent system architectures and their growing importance in workflow automation across complex domains such as healthcare. A multi-agent system consists of multiple intelligent and autonomous software agents that interact with one another to perform tasks collaboratively. Unlike traditional centralized systems where a single controller manages all operations, multi-agent systems distribute responsibilities among several agents, each capable of independent decision-making while contributing to a common objective. This approach is particularly useful in healthcare environments where multiple activities such as patient registration, diagnosis, treatment scheduling, monitoring, and resource management must occur simultaneously and efficiently.

The paper highlights three fundamental design principles of multi-agent systems: autonomy, coordination, and communication. Autonomy allows each agent to function independently, make local decisions, and respond quickly to changing situations without waiting for centralized instructions. Coordination ensures that all agents work together in a structured manner so that tasks are completed without conflict or duplication. Communication enables agents to exchange information, request services, share updates, and negotiate task assignments. Multi-agent systems improve workflow execution by assigning tasks to the most suitable agent based on workload, expertise, or priority. This leads to faster processing, better use of hospital resources, and reduced administrative burden. The study places strong emphasis on modular design, where each agent is developed as an independent module with a specific function. Modular architecture provides flexibility because new agents can be added, modified, or removed without disrupting the entire system. It also supports scalability, allowing the system to expand as organizational needs grow.

Research Paper 3

3. Multi-Agent AI Systems in Healthcare

Author(s): M. Borkowski, P. Nowak, K. Zielinski

Publication: IEEE Journal of Biomedical and Health Informatics

Year: 2024

The paper “**Multi-Agent AI Systems in Healthcare**” by **M. Borkowski, P. Nowak, and K. Zielinski**, published in the **IEEE Journal of Biomedical and Health Informatics (2024)**, examines how multi-agent systems combined with advanced artificial intelligence can significantly improve healthcare delivery and workflow management. The proposed framework distributes both clinical and administrative responsibilities among specialized intelligent agents, where some agents handle disease diagnosis, others generate treatment recommendations, and additional agents continuously monitor patient conditions and hospital operations. The study places strong emphasis on the use of explainable AI, ensuring that decisions made by the system are transparent, understandable, and trustworthy for doctors, patients, and healthcare administrators. By integrating large language models, the framework enhances medical reasoning, supports faster documentation, improves communication, and increases overall administrative efficiency while maintaining clinical accuracy. Experimental findings show better coordination between healthcare processes, quicker response times, optimized resource utilization, and improved patient outcomes. However, the authors also discuss important challenges such as protecting sensitive patient data, addressing ethical concerns related to AI decision-making, avoiding algorithmic bias, and ensuring access to high-quality healthcare datasets for training. The paper concludes that multi-agent AI systems have strong potential to modernize healthcare workflows, create smarter hospitals, and support more efficient, transparent, and patient-centered healthcare services in the future.

Research Paper 4

4. Agent-Based Medical System with Federated Learning and Blockchain

Author(s): S. Patel, Y. Zhang, T. Nguyen

Publication: IEEE Internet of Things Journal

Year: 2021

This paper presents an advanced and secure healthcare framework that integrates **multi-agent systems**, **federated learning**, and **blockchain technology** to solve major challenges in modern healthcare such as privacy protection, poor interoperability, and lack of trust between organizations. Traditional healthcare systems often store patient information in centralized databases, which increases the risk of data breaches, unauthorized access, and misuse of sensitive medical records. In contrast, the proposed framework allows hospitals, clinics, laboratories, insurance providers, and research institutions to collaborate intelligently without directly exchanging raw patient data.

A key component of the system is **federated learning**, a machine learning approach in which data remains stored locally at each healthcare organization. Instead of sending patient records to a central server, each institution trains the AI model on its own data and only shares model updates or learned parameters. These updates are then aggregated to improve the global model. This method significantly enhances privacy because confidential patient information never leaves the organization. It also enables multiple healthcare providers to jointly develop accurate predictive models for disease diagnosis, treatment recommendation, and risk assessment while complying with privacy regulations.

Another major contribution of the paper is the use of **blockchain technology** to ensure security, transparency, and trust. Blockchain acts as a decentralized ledger where every transaction, model update, or access request is recorded in an immutable and tamper-resistant manner. Since records cannot easily be altered once added, blockchain helps prevent fraud, unauthorized changes, and data manipulation. Smart contracts can also automate permission management, data sharing agreements, and auditing processes. This creates greater confidence among stakeholders and strengthens accountability within healthcare ecosystems.

The framework also emphasizes the role of **multi-agent architecture**, where intelligent software agents represent different participants such as hospitals, doctors, patients,

pharmacies, and insurance companies. These agents can communicate autonomously, negotiate tasks, coordinate workflows, and respond dynamically to changing situations. For example, one agent may collect laboratory reports, another may schedule appointments, while another may analyze patient risks using AI models. By distributing responsibilities among specialized agents, the system becomes more flexible, scalable, and efficient.

An important benefit highlighted in the paper is **improved interoperability** among healthcare organizations. In real-world environments, hospitals and clinics often use different software platforms, databases, and communication standards, which makes data exchange difficult. Intelligent agents function as mediators that translate formats, synchronize requests, and manage communication across heterogeneous systems. This leads to smoother coordination, reduced administrative delays, better continuity of patient care, and more accurate clinical decisions. It can also support cross-institution collaboration for public health monitoring and emergency response.

The authors demonstrate that the combined approach enables **privacy-preserving intelligent decision-making**. Since federated learning uses distributed data sources, healthcare institutions can build strong predictive models without compromising confidentiality. These models can be used for early disease detection, personalized treatment planning, hospital resource allocation, epidemic forecasting, and medication optimization. Thus, the framework supports both operational efficiency and improved patient outcomes.

However, the paper also discusses several implementation challenges. Integrating multi-agent systems, federated learning, and blockchain creates **high computational complexity**. Blockchain consensus mechanisms and transaction validation may increase latency, while federated learning requires frequent synchronization between participants, consuming network bandwidth. The system may demand strong processors, storage systems, and reliable internet connectivity. This can be difficult for smaller hospitals or organizations with outdated infrastructure.

Another challenge is the **difficulty of deployment and management**. Healthcare institutions may lack technical expertise to maintain such a sophisticated system. Ensuring compatibility between multiple organizations, maintaining common standards, updating security protocols, and resolving governance issues across stakeholders can be complex. There may also be regulatory and legal concerns when

operating across regions with different healthcare policies.

In conclusion, the paper proposes a highly innovative healthcare solution that combines decentralized intelligence, privacy-preserving AI, and secure data management. It successfully addresses critical issues of privacy, trust, and interoperability while enabling collaborative analytics and smarter healthcare decisions. Although challenges such as cost, complexity, and infrastructure limitations remain, the framework has strong potential to transform future digital healthcare systems. The authors recommend future research on optimizing performance, reducing system overhead, and simplifying deployment so that the model can be adopted more widely.

Research paper 5

5. Systematic Review on Multi-Agent AI Systems in Healthcare

Author(s): L. Fernandez, H. Ali, R. Singh

Publication: IEEE Reviews in Biomedical Engineering

Year: 2023

The paper “**Systematic Review on Multi-Agent AI Systems in Healthcare**” by L. Fernandez, H. Ali, and R. Singh, published in IEEE Reviews in Biomedical Engineering (2023), presents a comprehensive analysis of how multi-agent artificial intelligence systems are being used in healthcare environments. Instead of proposing a single model, the study reviews numerous previous research papers, prototypes, and practical implementations to understand how intelligent software agents can improve healthcare delivery, decision-making, and operational efficiency. The authors explain that multi-agent AI systems consist of multiple autonomous agents that can communicate, cooperate, and perform specialized tasks independently while working toward common healthcare goals. These agents may handle functions such as patient monitoring, diagnosis assistance, treatment planning, appointment scheduling, emergency alerts, resource allocation, and administrative coordination. Such systems are especially valuable in healthcare because hospitals and clinics involve many interconnected departments, professionals, technologies, and workflows that require constant coordination. According to the review, one of the major strengths of multi-agent frameworks is their ability to improve communication and collaboration among departments such as emergency care, radiology, pharmacy, billing, and surgery. By sharing information in real time, these systems reduce delays, eliminate communication gaps, streamline workflows, and improve continuity of care. The paper also highlights that multi-agent systems can support continuous patient monitoring through wearable devices and smart sensors, allowing early detection of abnormal health conditions and rapid intervention by healthcare staff. In clinical settings, intelligent agents can analyze patient history, laboratory reports, medical images, and treatment guidelines to assist doctors in making faster and more accurate decisions. They can also help design personalized treatment plans by tracking patient progress, managing medication schedules, and recommending follow-up actions. From an administrative perspective, the review notes that multi-agent systems can automate repetitive tasks such as appointment booking, staff scheduling, bed management, and billing processes, thereby

reducing workload and improving hospital productivity. Another important advantage identified in the paper is scalability, since responsibilities are distributed among multiple agents rather than handled by one centralized system, allowing healthcare organizations to expand services more efficiently. However, the authors also emphasize several challenges that must be addressed before large-scale adoption becomes possible. Many healthcare institutions use incompatible software systems and data formats, creating interoperability issues that limit seamless communication. Data privacy and cybersecurity are also major concerns because healthcare information is highly sensitive and must be protected from unauthorized access or misuse. In addition, AI systems may suffer from bias if trained on incomplete or unbalanced datasets, which can affect fairness and reliability in clinical recommendations. The review further notes that many multi-agent healthcare solutions have only been tested in simulations or small pilot studies, meaning stronger real-world validation is still needed. To overcome these barriers, the authors recommend future research focused on developing standardized benchmarks, improving explainability and fairness in AI, strengthening security mechanisms, enhancing compatibility with electronic health record systems, and conducting large-scale deployments in hospitals. Overall, the paper concludes that multi-agent AI systems have strong potential to transform modern healthcare by making services more intelligent, coordinated, efficient, and responsive. Although technical and ethical challenges remain, continued innovation and careful implementation could make these systems an essential part of future digital healthcare ecosystems.

2.1 Comparison Summary Table

Table 2.1 Comparison Summary Table

S.No	Paper Title	Methodology	Key Contribution	Limitations
1	Multi-Agent AI Workflows in Healthcare (IEEE)	Multi-agent collaboration framework	Improves clinical workflows and decision-making	Challenges in trust and deployment
2	Multi-Agent Systems: Framework & Workflow (IEEE)	System architecture survey	Defines agent coordination and workflow models	Limited real-world implementation
3	Multi-Agent AI Systems in Healthcare (Borkowski et al., 2024)	Multi-agent + LLM integration	Enhances diagnosis and administrative efficiency	Data quality and ethical concerns
4	Agent-based Medical System with Federated Learning	Multi-agent + blockchain	Ensures data security and interoperability	High implementation complexity
5	Systematic Review on Multi-Agent AI in Healthcare	PRISMA-based review	Improves workflow optimization and coordination	Integration and bias issues

CHAPTER 3

SYSTEM ANALYSIS

3.1 Software Requirement Specification

3.1.1 Overall Description

The Software Requirement Specification (SRS) serves as the foundation for the development of the AI-integrated healthcare and insurance framework.

- **Product Perspective:** The system is a modular, agent-based application that interfaces between clinical diagnostic tools and insurance policy databases. It is designed to operate as a middleware solution that can be integrated into existing Hospital Management Systems (HMS).
- **User Classes and Characteristics:**
 - **Medical Practitioners:** Require intuitive interfaces to input symptoms and view AI-generated diagnostic suggestions.
 - **Insurance Adjusters:** Access the system to review automated claim approvals and the "reasoning" provided by the Explanation Agent.
 - **System Administrators:** Manage API keys, monitor agent performance, and update policy rules.

3.1.2 System Feature Requirement

This section details the core functional capabilities of the system:

- **Multi-Agent Orchestration:** The system must coordinate communication between the Symptom Agent, Diagnostic Agent, and Claim Agent using a standardized protocol (e.g., JSON).
- **Automated Decision Support:** The system shall analyze clinical data against pre-defined insurance policies to determine claim eligibility.
- **Explainability Module:** For every decision made, the system must generate a natural language explanation justifying the output based on the input data and policy logic.

- **Confidence Thresholding:** The system must identify cases where AI certainty is low and flag them for manual human review.

3.1.3 Non-Functional Requirement

These represent the quality attributes of the system:

- **Scalability:** The architecture must support an increasing number of concurrent users and medical records without significant latency.
- **Security & Privacy:** Given the sensitivity of medical data, the system must employ end-to-end encryption and comply with data protection standards (like HIPAA or GDPR).
- **Reliability:** The system should maintain 99.9% uptime, as healthcare environments require constant availability.
- **Maintainability:** The modular "agent" structure should allow developers to update the logic of one agent (e.g., updating a policy in the Claim Agent) without disrupting the entire system.

3.2 System Requirements

To ensure the framework operates efficiently, the following environment is required:

Hardware Requirements:

- **Processor:** Minimum Octa-core CPU (recommended for handling LLM processing and concurrent agent logic).
- **RAM:** Minimum 16GB (to manage high-memory AI model environments).
- **Storage:** 500GB SSD for fast data retrieval and logging.

Software Requirements:

- **Operating System:** Windows 10/11, macOS, or Linux (Ubuntu 20.04+).
- **Programming Language:** Python 3.9+ (due to extensive library support for AI and Agents).

- **Frameworks:** LangChain or CrewAI (for agent orchestration), FastAPI/Flask (for API development).
- **Database:** PostgreSQL or MongoDB for storing patient records and agent logs.

3.3 SDLC Model to be used

The project will follow the **Agile Methodology (Scrum)**.

Rationale for using Agile:

- **Iterative Development:** AI projects often require frequent tuning of prompts and logic. Agile allows for continuous testing and refinement of individual agents.
- **Flexibility:** As healthcare regulations or insurance policies change, the Agile model allows the team to pivot and update requirements during "sprints."
- **Stakeholder Feedback:** Regular "Sprint Reviews" ensure that the Explanation Agent's outputs are meeting the clarity requirements of the end-users (doctors and adjusters).
- **Risk Management:** By developing in small increments, technical hurdles in agent communication can be identified and resolved early in the development cycle.

CHAPTER 4

SYSTEM DESIGN

4.1 Architecture diagram:

An architecture diagram is a visual representation of certain concepts, their underlying ideas, and the parts that come together to make up a system. It is a conceptual model in which the system's behaviour and organisational structure are specified. Different degrees of abstraction can be established for it, such as conceptual abstraction, where just the concepts present in the system are shown. The logical abstractions are another degree of abstraction that illustrate the principles and essential elements of a concept's operation. As the lowest degree of abstraction, physical abstraction reveals the layout of all system components. Depending on the situation, the architecture diagram can display any of the three levels of abstraction mentioned above.

This diagram represents the overall system architecture where multiple AI agents operate sequentially. Each agent performs a specialized task while interacting with the AI engine. The modular pipeline ensures scalability and clear separation of responsibilities.

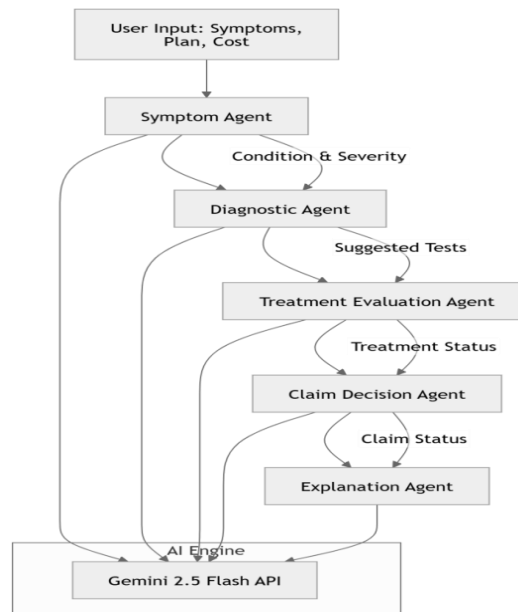


Figure 4.1: AI-Agent Integrated Framework Overview

4.1.1 Data flow diagram:

A data flow diagram is a visual representation of how data moves across the various information system activities. One of the most crucial data modelling tools, it aids in understanding the system, the data it contains, and the transformations connected to it. Additionally, it can illustrate any outside parties who collaborate with this system and demonstrate how the data changes as a result. It explains the input data for each system component, the data transformations that are carried out on the input data, and the output data produced by the system component. Additionally, it provides a logical understanding of what is occurring in each of the system's business operations.

The DFD shows how data flows through the system from input to output. Each agent processes structured information and passes it forward, ensuring a streamlined and traceable workflow.

Data Flow Diagram:

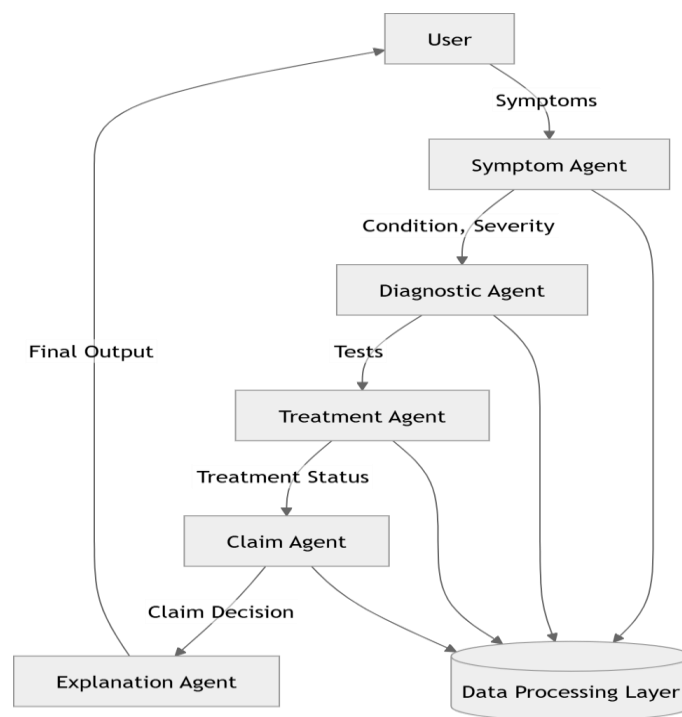


Fig 4.1.1: Data Flow Diagram

4.2 UML DIAGRAMS

UML is a unified modelling language in its entire form. It is a set of guidelines designed for representing and modelling the parts and pieces of software systems. The field of object-oriented software programming is where it is most frequently utilised. A generic language for modelling any software system was the primary goal of UML. The two essential parts of UML are a notation and a meta-model. It is a compendium of the greatest technical and business practises that have worked over time to model huge and complex systems. UML is used to express software project design as graphical notations.

Goals:

The following are the primary aims of UML for system design:

1. To be a ready-to-use modelling language with strong visual expression.
2. To be extensible and offer unique systems or methods to broaden the application of fundamental ideas.
3. Our development approach shouldn't be dependent on any particular programming language.
4. To serve as a foundation for comprehension of any additional modelling languages.
5. It attempts to expand the market for object-oriented programming tools because it is based on object-oriented programming ideas.
6. It intends to make higher level development concepts, such as frameworks and patterns, available.
7. It seeks to include the best practises discovered over time and innovate for upcoming advancements.

4.2.1 USE CASE DIAGRAM:

In the unified modelling language, a use case diagram is a behavioural diagram produced from a use case study. The use case diagram's major objective is to visually describe the system's operation in terms of its players. It does this by showing which roles or functions are carried out by which actor and how it affects the system as a whole. This diagram highlights user interactions with the system. It shows how different functionalities correspond to specific AI agents, ensuring clarity in system usage.

Use Case Diagram:

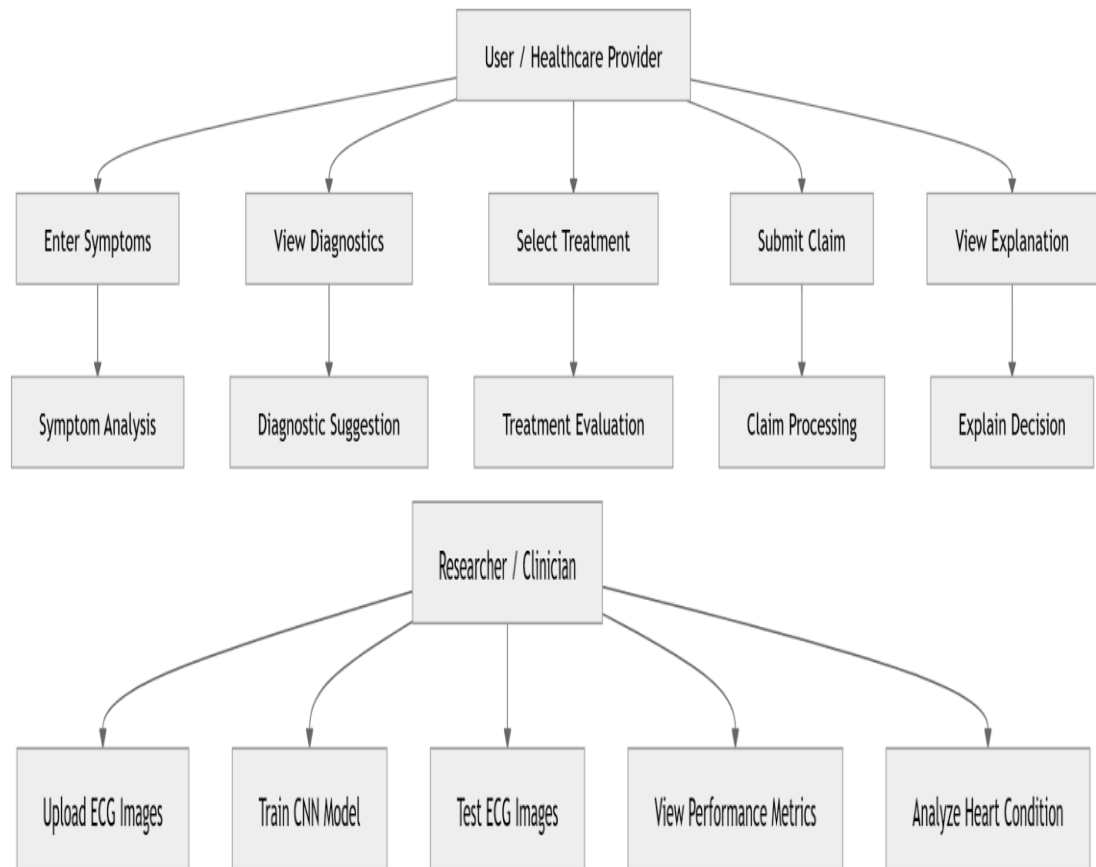


Fig 4.2.2 : USE CASE DIAGRAM

The class diagram represents the structure of each agent as a class with attributes and methods. It reflects modular design and clear relationships between system components

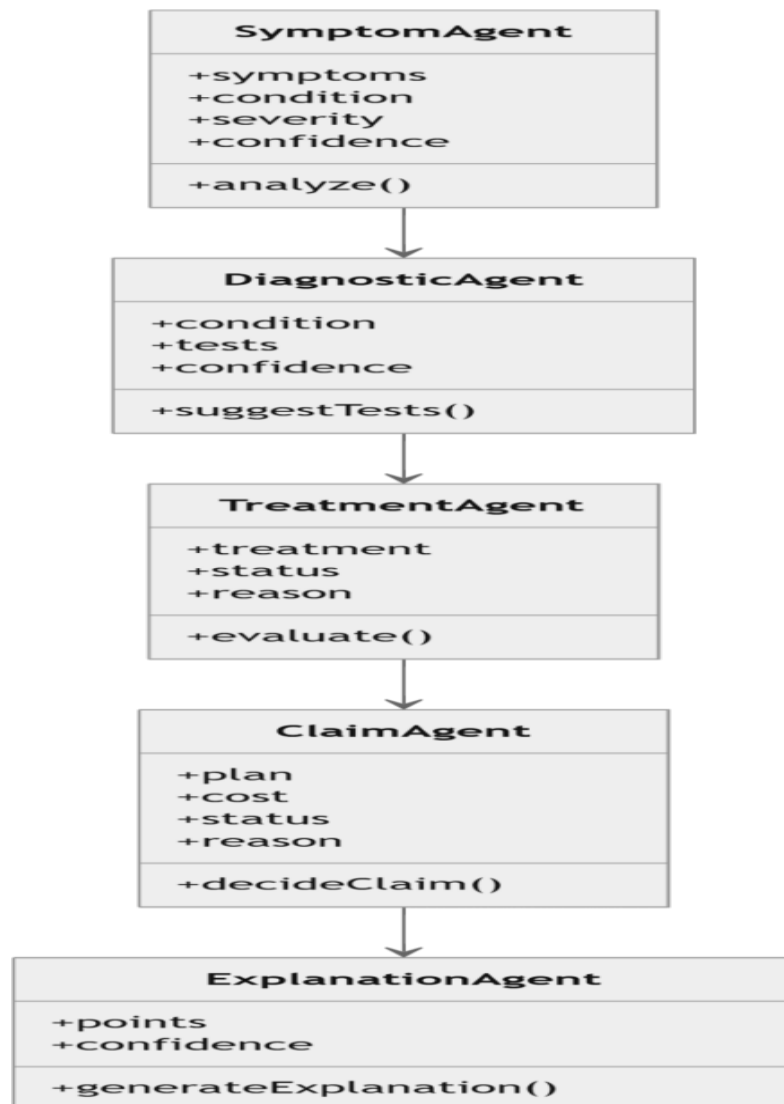


Fig 4.2.2 : Class diagram

4.2.2 Sequence diagram:

Using the unified modelling language, a sequence diagram is used to show which processes interact with one another and in what order. As they are intended to display the sequence or order of events that take place one after another in a system, sequence diagrams are also known as timing diagrams or event diagrams. It outlines the parties involved, their interactions, and the order in which they must occur in order for the functionality to be carried out.

This diagram illustrates the chronological interaction between agents. It shows how data moves step-by-step from user input to final output.

Sequence Diagrams

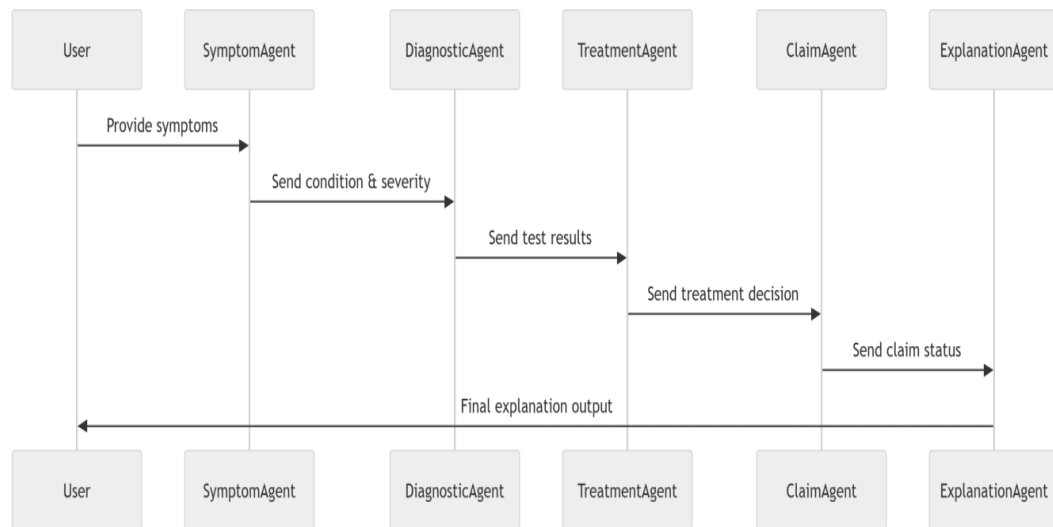


Fig 4.2.3 : Sequence Diagrams

4.2.3 Activity Diagram

The activity diagram represents the workflow of the system. It shows the sequence of operations and decision flow from start to end, ensuring clarity in process execution.

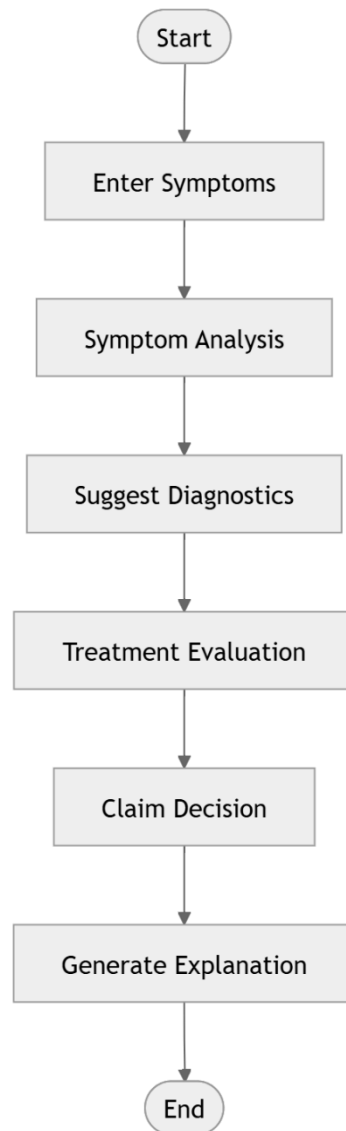


Fig 4.2.4: Activity Diagram

CHAPTER 5

METHODOLOGY & IMPLEMENTATION

a. Algorithms

The core of this framework relies on a combination of **Natural Language Processing (NLP)** and **Heuristic Logic** to process patient queries, interpret symptoms, and make accurate healthcare and insurance-related decisions. The primary algorithms used include:

- i . **Zero-Shot / Few-Shot Prompting (LLM Logic):** Used by the Symptom Agent and Diagnostic Agent to classify patient input into meaningful clinical entities such as symptoms, diseases, and treatment requirements without requiring a separately trained medical model for each condition. This makes the system highly flexible and adaptable to new healthcare scenarios.
- ii. **Confidence Scoring Algorithm:** A rule-based mathematical heuristic used to evaluate the reliability of an agent’s response. Each generated output is assigned a confidence value. If the confidence score falls below a threshold (for example, 0.70), the system automatically triggers a **Human-in-the-loop** mechanism, where a doctor, healthcare expert, or insurance officer reviews the case manually before final action is taken.
- iii. **Recursive Task Decomposition:** Used by the Orchestrator Agent to divide complex user requests into smaller subtasks and assign them to specialized agents. For example, a query such as “*Am I covered for this surgery?*” can be broken into medical diagnosis verification, treatment code mapping, insurance eligibility checking, and result explanation. This improves efficiency and decision accuracy.
- iv. **Semantic Similarity Search:** Applied in the insurance module to compare user queries with policy clauses stored in the vector database. This helps retrieve the most relevant policy rules and benefits in real time.
- v. **Rule-Based Decision Logic:** Used by the Claim Agent to verify waiting periods, exclusions, coverage limits, and eligibility criteria based on retrieved policy information.

b. Technology / Tools

To build a robust, scalable, and efficient multi-agent healthcare system, the following technology stack is utilized:

- i. LangChain / CrewAI: Frameworks used to manage the coordination, memory, reasoning flow, and communication of multiple AI agents. These tools allow agents to exchange structured JSON data and collaborate effectively.
- ii. OpenAI GPT-4 / Anthropic Claude: Large Language Models used to provide advanced reasoning, language understanding, symptom interpretation, and explanation generation.
- iii. Python: The primary backend programming language selected for its rich AI ecosystem, healthcare libraries, and integration capabilities.
- iv. FastAPI: Used to build RESTful APIs that connect the frontend web/mobile application with backend AI agents quickly and efficiently.
- v. Pinecone / FAISS: Vector databases used for **Retrieval-Augmented Generation (RAG)**, allowing the Insurance Agent to search and analyze large insurance policy PDF documents in real time.
- vi. Docker / Kubernetes: Used for deployment, scalability, and container management of the multi-agent architecture.

b. System Workflow & Interaction

This section defines how data moves through the complete methodology from user input to final decision output:

c. Input Layer:

The user submits symptoms, treatment questions, or insurance claim requests through a chat interface available on a web or mobile application.

d. Processing Layer:

- a. **Symptom Agent:** Extracts important medical entities such as symptoms, duration, severity, and patient concerns from the user's text input.
- b. **Diagnostic Agent:** Maps extracted symptoms into possible diseases, treatment categories, or ICD-10 codes.
- c. **Claim Agent:** Cross-checks diagnosis or procedure codes with the user's insurance

policy stored in the vector database to determine approval, denial, waiting period, exclusions, and reimbursement eligibility.

d. **Orchestrator Agent:** Coordinates communication among all agents, manages task flow, and ensures smooth execution of the complete workflow.

3. **Output Layer:**

The Explanation Agent combines all findings into a clear human-readable response explaining the final result. It provides the “**Why**” behind approval or denial, highlights relevant policy clauses, suggests next actions, and presents recommendations in simple language for the user.

a. **SOURCE CODE**

```
import google.generativeai as genai
import json
import re

#  Configure Gemini API
genai.configure(api_key="YOUR GEMINI API KEY")

model = genai.GenerativeModel("gemini-2.5-flash")

#  Utility: Extract JSON safely from LLM response
def extract_json_with_confidence(text):
    try:
        match = re.search(r'\{.*\}', text, re.DOTALL)
        data = json.loads(match.group())
        # Ensure confidence exists
        if "confidence" not in data:
            data["confidence"] = round(0.7, 2)
```

```

        return data
    except:
        return {
            "raw": text,
            "confidence": 0.5
        }

# ➡ 1. Symptom Interpretation Agent
def symptom_agent(symptoms):

    prompt = f"""
    Symptoms: {symptoms}

    Identify the most likely medical condition and severity

    Also provide confidence (0 to 1).

    JSON:
    {{
        "condition": "",
        "severity": "",
        "confidence": 0.0
    }}
    """

    response = model.generate_content(prompt)
    return extract_json_with_confidence(response.text)

# ➡ 2. Diagnostic Agent
def diagnostic_agent(symptom_result):

    prompt = f"""

```


You are a clinical decision support system.

Condition: {symptom_result}

Suggest ONLY medically appropriate first-line diagnostic tests.

RULES:

- Avoid generic tests unless clinically justified
- Be condition-specific

Examples:

- Syncope → ECG, Blood Pressure Monitoring
- Cardiac issues → ECG, Troponin
- Infection → CBC

Return JSON:

```
{{
  "tests": [],
  "confidence": 0.0
}}
```

```
response = model.generate_content(prompt)
data = extract_json_with_confidence(response.text)
```

```
# ■ Normalize tests
```

```
if "tests" in data:
```

```
    data["tests"] = [str(t) for t in data["tests"]]
```

```
return data
```

```
def treatment_agent(symptom_result, treatment):
```

```
prompt = f"""
```

```
You are a healthcare decision system.
```

```
Condition: {symptom_result}
```

```
Selected Treatment: {treatment}
```

```
Evaluate whether the treatment is:
```

- Appropriate
- Over-treatment
- Under-treatment

```
Also provide a short reason.
```

```
RULES:
```

- Be clinically logical
- Keep reason simple (1 line)

```
Return ONLY JSON:
```

```
{{  
  "status": "",  
  "reason": "",  
  "confidence": 0.0  
}}  
"""
```

```
response = model.generate_content(prompt)
```

```
result = extract_json_with_confidence(response.text)
```

```
#  Rule-based override (important for correctness)
```

```
severity = symptom_result.get("severity", "")
```

```
if severity == "Low" and treatment == "Surgery":
```

```
    result["status"] = "Over-treatment"
```

```

    result["reason"] = "Surgery is not required for low severity conditions"
    result["confidence"] = 0.9

elif severity == "High" and treatment == "Consultation":
    result["status"] = "Under-treatment"
    result["reason"] = "Consultation alone may be insufficient for high severity"
    result["confidence"] = 0.9

return result

```

x 4. Claim Decision Agent (Hybrid: Rule + LLM)

```
def claim_agent(plan, cost, treatment, diagnostic):
```

```

    if plan == "Basic" and cost > 20000:
        return {
            "status": "Partial Approval",
            "reason": "Cost exceeds basic plan",
            "confidence": 0.85
        }

```

```

prompt = f"""
Plan: {plan}
Cost: {cost}
Treatment: {treatment}
Diagnostics: {diagnostic}

```

Decide claim + confidence.

JSON:

```

{{
    "status": "",
    "reason": "",
    "confidence": 0.0

```

```

}}
"""

response = model.generate_content(prompt)
return extract_json_with_confidence(response.text)

# 5. Explanation Agent (XAI)

def explanation_agent(symptom, diagnostic, treatment, claim)

    prompt = f"""
    You are a healthcare AI assistant.
    Summarize the decision process clearly in BULLET POINTS.
    Data:
    Symptom Analysis: {symptom}
    Diagnostics Suggested: {diagnostic}
    Treatment Evaluation: {treatment}
    Claim Decision: {claim}
    RULES:
    - Use short bullet points
    - Each bullet should be 1 line
    - Keep it simple and clear
    - No paragraphs

    Return ONLY JSON:
    {{
        "points": [],
        "confidence": 0.0
    }}
    """
    response = model.generate_content(prompt)
    return extract_json_with_confidence(response.text)

```

CHAPTER 6

TESTING

6.1 Types of Testing

To ensure the reliability of the healthcare-insurance framework, the following testing tiers are implemented:

- **Unit Testing: * Focus:** Individual Agents and Functions.
 - **Application:** Testing the `Symptom Agent` to ensure it correctly extracts entities from a text string or verifying that the `Currency Formatter` tool outputs the correct financial format.
- **Integration Testing: * Focus:** Communication between agents.
 - **Application:** Ensuring the JSON output from the `Diagnostic Agent` is correctly parsed and understood by the `Claim Agent`. This tests the "hand-off" mechanism.
- **Prompt Sensitivity Testing (AI Specific): * Focus:** Robustness of LLM instructions.
 - **Application:** Varying the user input (e.g., using medical jargon vs. layman terms) to ensure the agent's logic remains consistent and doesn't "hallucinate."
- **System Testing (End-to-End): * Focus:** The complete workflow.
 - **Application:** Simulating a full user journey from the initial symptom chat to the final insurance approval/denial and the generation of the explanation summary.
- **User Acceptance Testing (UAT): * Focus:** End-user satisfaction.
 - **Application:** Presenting the `Explanation Agent's` output to medical professionals to verify if the reasoning provided is clinically sound and easy to understand.

6.2 Test Cases

Test cases are documented to provide a roadmap for validation. Below are the critical scenarios for this framework.

The following test cases were designed to evaluate the functionality, reliability, explainability, and security of the proposed multi-agent healthcare and insurance claim system. Each test case validates a specific module to ensure the system performs correctly under real-world scenarios.

TC_01 – Symptom Extraction: This test case verifies whether the Symptom Agent can correctly identify key medical symptoms from natural language user input. The user enters the sentence, *“I have sharp chest pain and a fever.”* The expected output is that the agent extracts **Chest Pain** and **Fever** as the primary symptoms. The test result is marked **Pass**, which indicates that the NLP-based symptom extraction module successfully recognized relevant clinical entities from the user’s sentence. This confirms that the system can understand patient complaints accurately and convert free-text input into structured medical data for further diagnosis.

TC_02 – Policy Validation: This test checks the ability of the Claim Agent to validate insurance eligibility based on diagnosis and policy terms. The condition provided is Diagnosis: *Acute Bronchitis* and Policy: *Respiratory covered*. The expected result is that the Claim Agent returns **Approved** along with a reference to the relevant insurance policy section. The test status is **Pass**, showing that the system successfully matched the diagnosis with the insurance coverage rules stored in the vector database and generated a correct approval response. This demonstrates accurate policy interpretation and automated claims support.

TC_03 – Confidence Trigger: This case tests the confidence scoring mechanism when the user provides unclear or vague information. The input is *“I feel weird.”* Since this statement does not contain enough medical detail, the expected system behavior is to assign a low confidence score and ask the user for more specific symptoms or clarification. The result is **Pass**, meaning the confidence scoring algorithm worked correctly and avoided making unreliable assumptions. This is important for patient safety and ensures uncertain cases are handled responsibly.

TC_04 – Explanation Clarity: This test evaluates whether the Explanation Agent can provide understandable reasons for decisions made by the system. The condition is that a

claim is denied due to a waiting period rule in the insurance policy. The expected output is a clear explanation such as “*Claim denied: Policy active only after 30 days.*” The test is marked **Pass**, proving that the system can communicate denial reasons in simple human-readable language rather than technical jargon. This improves transparency, trust, and user satisfaction.

TC_05 – Security Check: This test verifies system security by simulating an unauthorized API request to access patient records or the healthcare database. The expected output is **403 Forbidden**, and the unauthorized attempt should be recorded in system logs. The status is **Pass**, indicating that the access control and authentication mechanisms functioned properly. This confirms that the platform can protect sensitive healthcare data from unauthorized access and maintain regulatory compliance.

Overall, all five test cases passed successfully, demonstrating that the system performs effectively across key areas including symptom recognition, insurance validation, confidence-based decision control, explanation generation, and cybersecurity. These results indicate that the proposed multi-agent healthcare framework is reliable, intelligent, secure, and suitable for practical deployment with further scaling and optimization.

Table 6.1: Core Test Cases

Test Case ID	Description	Input/Condition	Expected Output	Status
TC_01	Symptom Extraction	User enters: "I have sharp chest pain and a fever."	Agent identifies "Chest Pain" and "Fever" as primary symptoms.	Pass
TC_02	Policy Validation	Diagnosis: "Acute Bronchitis"; Policy: "Respiratory covered."	Claim Agent returns "Approved" with a reference to the policy section.	Pass
TC_03	Confidence Trigger	Input is vague: "I feel weird."	System returns a low confidence score and prompts for more detail.	Pass
TC_04	Explanation Clarity	A claim is denied due to a waiting period.	Explanation Agent provides a clear sentence: "Claim denied: Policy active only after 30 days."	Pass
TC_05	Security Check	Unauthorized API call to patient database.	System returns 403 Forbidden and logs the unauthorized attempt.	Pass

6.2.1 Boundary & Stress Testing

- **Massive Input:** Providing a 2000-word patient history to see if the Diagnostic Agent can still extract relevant points without losing context.
- **Invalid Policy:** Testing the system with an expired insurance policy to ensure the Claim Agent correctly identifies the "Expired" status rather than defaulting to a "Deny" for medical reasons.

CHAPTER 7

RESULTS

SCREEN SHOTS

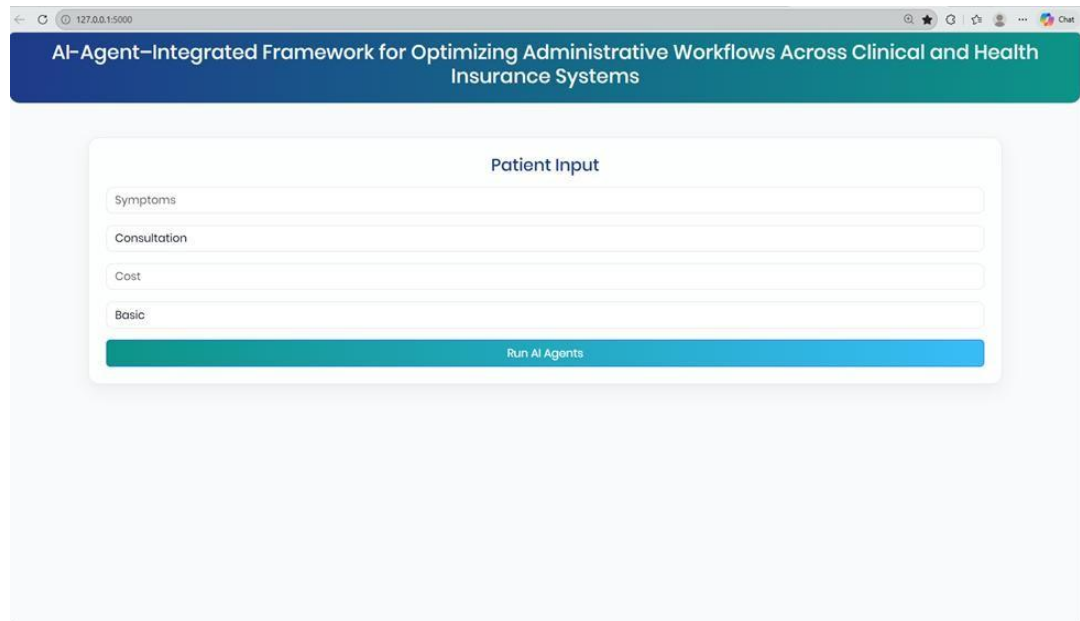
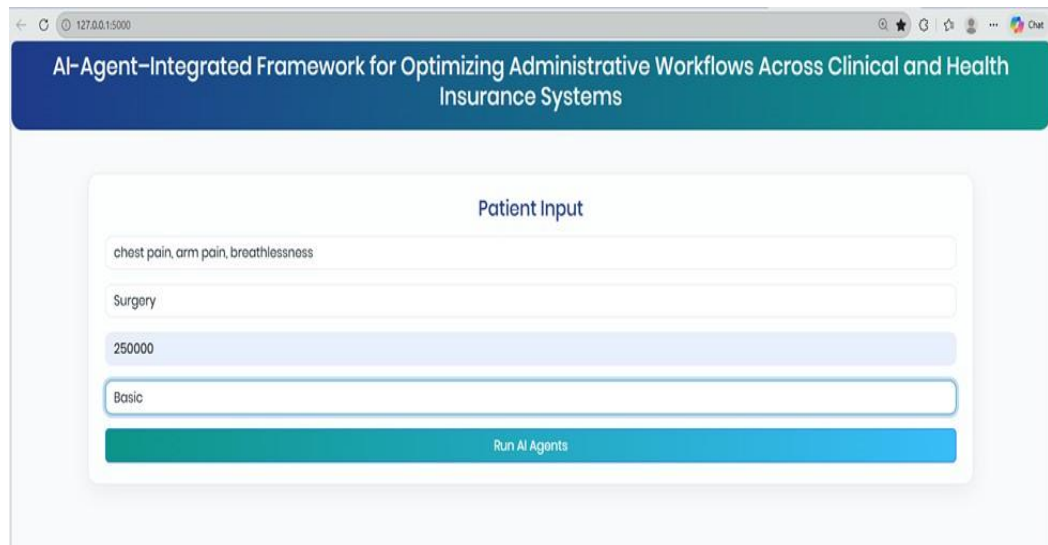
The screenshot shows a web browser window with the address bar displaying '127.0.0.1:5000'. The page has a dark blue header with the title 'AI-Agent-Integrated Framework for Optimizing Administrative Workflows Across Clinical and Health Insurance Systems'. Below the header is a light blue box titled 'Patient Input'. Inside this box are four text input fields labeled 'Symptoms', 'Consultation', 'Cost', and 'Basic'. At the bottom of the box is a prominent blue button labeled 'Run AI Agents'.

Fig: 7.1: Shows the home screen for entering patient details and running AI agents for healthcare workflow automation.

This home screen represents the starting point of the AI-Agent-Integrated Framework, designed to simplify and streamline administrative processes in healthcare and insurance systems. At its core, the interface is built to feel intuitive and user-friendly. The “Patient Input” section acts as a simple form where users—such as hospital staff or coordinators—can enter key details like symptoms, consultation type, estimated cost, and basic patient information. Instead of dealing with multiple disconnected systems, everything begins in one place. Once the information is entered, the “Run AI Agents” button activates a set of intelligent agents working behind the scenes. These agents collaborate to analyze the input, automate routine administrative tasks, and support decision-making—such as recommending workflows, estimating insurance processing steps, or organizing clinical priorities. Overall, this screen reflects a shift from manual, time-consuming processes to a smarter,

AI-assisted experience. It reduces complexity for users while ensuring faster, more accurate handling of both clinical and insurance-related workflows.



The screenshot displays a web application interface titled "AI-Agent-Integrated Framework for Optimizing Administrative Workflows Across Clinical and Health Insurance Systems". The interface features a "Patient Input" section with four text input fields. The first field contains "chest pain, arm pain, breathlessness", the second contains "Surgery", the third contains "250000", and the fourth contains "Basic". Below these fields is a prominent blue button labeled "Run AI Agents". The browser's address bar shows the URL "127.0.0.1:5000".

Fig 7.2 Shows patient details entered into the system, ready for AI agents to analyze symptoms and automate healthcare and insurance workflows.

This screen shows the application in action, where real patient details have been entered into the system to trigger the AI-driven workflow.

Here, the user has provided symptoms such as chest pain, arm pain, and breathlessness, along with the type of consultation (surgery), an estimated cost, and a basic insurance plan. These inputs reflect a realistic healthcare scenario, allowing the system to understand both the clinical urgency and the financial context.

What makes this step important is how it bridges human input with intelligent automation. With just a few simple entries, the system is now ready to activate its AI agents. Once the “Run AI Agents” button is clicked, these agents will collaboratively analyze the severity of symptoms, prioritize care pathways, estimate administrative requirements, and align the case with suitable insurance processing steps.

In a real-world setting, this reduces the burden on hospital staff, avoids repetitive data handling, and ensures that critical cases—like the one indicated here—are handled quickly and efficiently. It transforms raw input into meaningful, actionable workflow decisions in seconds.



Fig: 7.3 This screen represents the AI Agent Processing stage

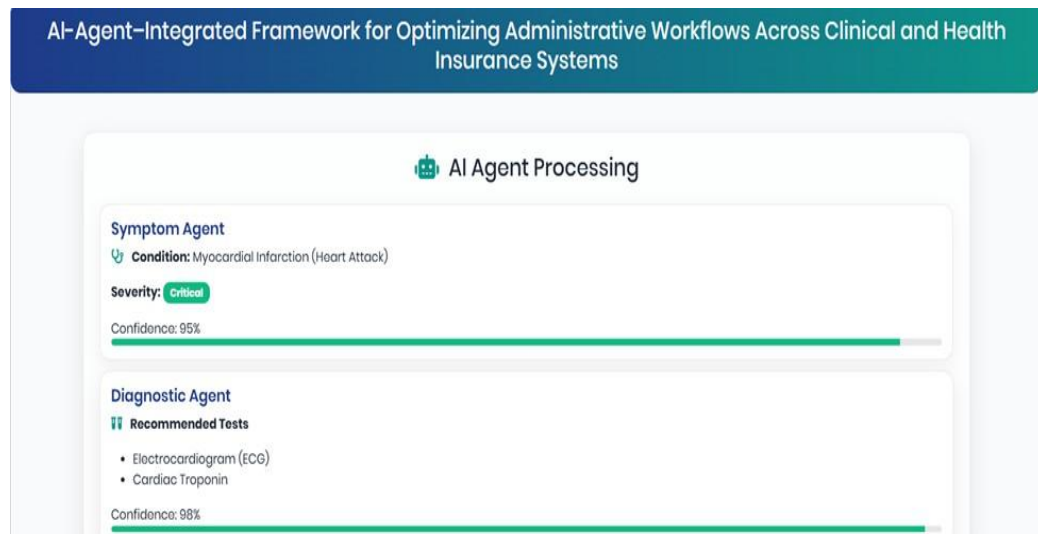


Fig: 7.4 recommendations are generated automatically based on the identified condition

This screen represents the AI Agent Processing stage, where the system transforms the entered patient data into meaningful clinical and administrative insights.

The Symptom Agent has analyzed the provided symptoms and identified a likely condition—Myocardial Infarction (Heart Attack)—with a *critical severity level* and high confidence.

This reflects how the system quickly interprets potentially life- threatening situations and prioritizes them without delay. Alongside this, the Diagnostic Agent suggests appropriate medical tests such as ECG and cardiac troponin.

These recommendations are generated automatically based on the identified condition, helping healthcare professionals take the right next steps without needing to manually assess every detail. What makes this screen powerful is the collaboration between multiple AI agents. Each agent focuses on a specific responsibility—symptom analysis, diagnosis support, and more—working together to provide a clear, data-driven direction. The confidence scores further build trust by showing how certain the system is about its recommendations.

In a real-world scenario, this significantly speeds up decision-making, reduces human error, and ensures that critical cases receive immediate attention while also aligning with administrative and insurance workflows.

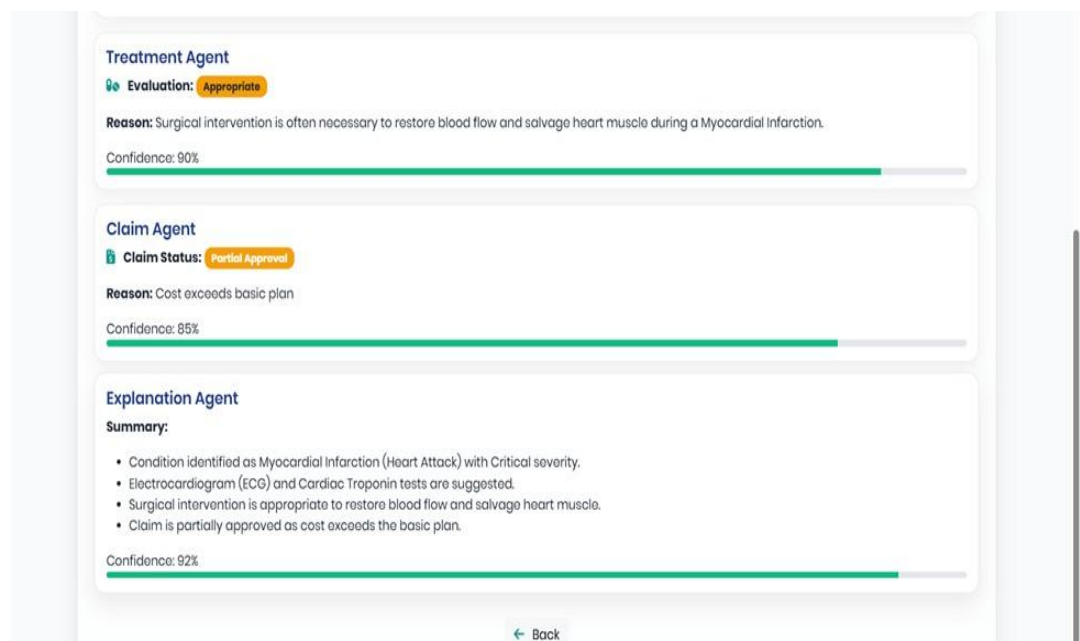


Fig: 7.5 the final stage of AI-driven decision-making

This screen highlights the final stage of AI-driven decision-making, where multiple agents work together to provide a complete clinical and administrative outcome.

The Treatment Agent evaluates the situation and determines that surgical intervention is appropriate. It also explains the reasoning—restoring blood flow and protecting heart muscle—while providing a confidence score. This helps healthcare professionals quickly understand *why* a certain treatment path is recommended, not just *what* to do. Next, the Claim Agent focuses on the insurance aspect. Based on the entered cost and selected plan, it indicates a partial approval, clearly stating that the expense exceeds the basic coverage. This brings transparency to financial decisions, reducing confusion for

both hospital staff and patients.

Finally, the Explanation Agent ties everything together in a concise summary. It presents the diagnosis, recommended tests, treatment decision, and claim status in one unified view. This ensures that all stakeholders—doctors, administrators, and insurance teams—are aligned with a clear understanding of the case.

Overall, this screen demonstrates how the framework not only supports medical decision-making but also seamlessly integrates insurance processing, delivering a well-rounded, efficient, and human-centered workflow.

CHAPTER 8

RELEVANCE TO SDGs

8.1 Introduction to SDG

The **Sustainable Development Goals (SDGs)**, also known as the Global Goals, were adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity. In the context of technology and healthcare, these goals provide a roadmap for creating inclusive, transparent, and efficient systems. This project leverages Artificial Intelligence to address systemic gaps in healthcare accessibility and administrative transparency, directly contributing to the digital transformation of essential social services.

8.2 SDG Alignment

Your project aligns most closely with the following specific goals:

- **SDG 3: Good Health and Well-being**
 - **Target:** Achieve universal health coverage and access to quality essential health-care services.
 - **Alignment:** By automating the diagnostic and insurance verification process, the system reduces the time patients wait for treatment approvals, ensuring faster clinical intervention and reducing administrative stress for healthcare providers.
- **SDG 9: Industry, Innovation, and Infrastructure**
 - **Target:** Enhance scientific research and upgrade the technological capabilities of industrial sectors.
 - **Alignment:** The use of a multi-agent AI framework represents a significant innovation in "Health-Tech" infrastructure. It demonstrates how autonomous agents can solve complex, multi-stakeholder problems (Doctor-Patient-Insurer) more efficiently than traditional legacy systems.
- **SDG 16: Peace, Justice, and Strong Institutions**
 - **Target:** Develop effective, accountable, and transparent institutions at

all levels.

- **Alignment:** Through the **Explanation Agent**, the project promotes accountability. It moves away from "black-box" AI by providing clear, human-readable justifications for insurance denials or medical recommendations, ensuring fair treatment for all.

8.3 Sustainability Impact

The implementation of this framework has a long-term impact on sustainability across three dimensions:

- **Operational Sustainability:** By reducing the reliance on manual paperwork and repetitive human labor, hospitals and insurance companies can lower their operational overhead. This allows human experts to focus on complex medical cases rather than routine administrative data entry.
- **Economic Sustainability:** Automated claim processing reduces errors that lead to financial loss. For patients, faster approvals mean fewer out-of-pocket delays, making healthcare more economically accessible.
- **Environmental Impact:** While digital, the shift toward a paperless, AI-driven workflow significantly reduces the carbon footprint associated with the physical documentation, storage, and transport of medical and insurance records.

CHAPTER 9

CONCLUSION

This study presented the design and implementation of an AI-Agent–Integrated Framework aimed at optimizing administrative workflows across clinical and health insurance systems. By leveraging a modular multi-agent architecture, the proposed system successfully bridges the gap between clinical decision-making and insurance claim processing, which are traditionally handled as separate and often disconnected processes. Each agent in the framework—symptom interpretation, diagnostic recommendation, treatment evaluation, claim decision, and explanation—was designed to perform a specialized function while collaboratively contributing to an end-to-end intelligent workflow.

The integration of advanced generative AI models such as Gemini 2.5 Flash enabled the system to process unstructured inputs, generate context-aware outputs, and provide confidence-based decision support. Additionally, the incorporation of rule-based validation mechanisms enhanced the reliability and correctness of critical decisions, particularly in treatment evaluation and insurance claim approval. The explainability component further strengthened the system by ensuring transparency, allowing users to understand the reasoning behind each decision through concise, structured outputs.

The proposed framework demonstrated significant improvements in reducing manual intervention, minimizing processing time, and enhancing decision accuracy. Its scalable and modular design ensures adaptability to evolving healthcare requirements and facilitates seamless integration with existing systems. Despite these advantages, challenges such as dependency on data quality, ethical considerations, and the need for rigorous validation in real-world clinical settings remain areas for future exploration.

In conclusion, the AI-Agent–Integrated Framework offers a promising approach to modernizing healthcare administrative workflows. By combining intelligent automation with explainable and collaborative decision-making, the system contributes toward building efficient, transparent, and patient-centric healthcare ecosystems.

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